# From Point Estimates to Hypocenter Distributions: A Controlled Comparison in 3D Joint Traveltime Tomography

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# 1 Introduction

Local-earthquake travel-time tomography uses picked seismic arrival times to estimate three-dimensional velocity structure and earthquake locations. Early work already formulated the inversion in terms of both source and medium parameters (Aki & Lee, 1976). An earthquake hypocenter is not an independently measured datum: its location is inferred from arrival times using an assumed velocity model, while velocity reconstruction in turn depends on the event locations (Kissling, 1988; Thurber, 1992). This coupling motivates estimating earthquake locations and velocity structure together.

Joint inversion of earthquake arrival times for three-dimensional velocity structure and hypocentral parameters has a long methodological history, with established implementations including the SIMUL family (Thurber, 1983), tomoDD (Zhang & Thurber, 2003), and LOTOS (Koulakov, 2009). More recent developments include eikonal-equation-based adjoint-state methods for jointly updating velocity structure and earthquake locations (Tong et al., 2024). Within this broader class of coupled inverse problems, probabilistic approaches account for uncertainty in source parameters, either together with uncertainty in velocity structure or conditional on an updated point estimate of the velocity model.

Existing comparisons within this probabilistic class provide evidence that accounting for source uncertainty can improve velocity reconstruction, but differ in how directly they test the benefit of hypocenter distributions over point estimates. The closest comparison is DeepGEM: Gao et al. (2021) retain conditional source-location distributions while updating a point estimate of the velocity model through variational expectation–maximization, and obtain better velocity reconstructions than joint maximum a posteriori estimation in two-dimensional synthetic experiments. However, the optimization procedure also differs between the two formulations. In three-dimensional synthetic tests, Agata et al. (2026) reduce velocity artifacts by analytically marginalizing uncertain source locations and origin times under Gaussian and linearization assumptions, compared with treating erroneous source parameters as fixed; this demonstrates the importance of source uncertainty, but does not compare against iterative point-estimate relocation and also estimates a velocity posterior. Yang et al. (2025) jointly infer velocity and source parameters through variational double-difference tomography, reporting more accurate synthetic

velocity models than tomoDD together with uncertainty estimates, although this comparison also changes the treatment of velocity uncertainty and regularization. Other comparisons show agreement with conventional inversions rather than clearly distinct velocity reconstructions: Piana Agostinetti et al. (2015) find that differences between their trans-dimensional three-dimensional Bayesian reconstruction and SimulPS are generally within posterior uncertainty, while Ryberg and Haberland (2019) obtain good agreement between Bayesian and conventional joint inversion of field data for one-dimensional velocity structure. These studies establish the potential value of probabilistic source treatment, but their comparisons do not by themselves isolate the effect of retaining hypocenter distributions rather than repeatedly updating point locations under otherwise matched three-dimensional inversion settings.

# 2 Method

# 3 Experimental Setup

# 4 Results

# 5 Discussion

# 6 Conclusion

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